

4. Focus Area 4 – Private Astronaut Missions to the International Space Station (ISS)

Focus Area 4A – Private Astronaut Mission (PAM) Provider

NASA has outlined a broad strategy to facilitate the commercialization of low Earth orbit (LEO) by U.S. companies (see [NID 8600.121: “Use of International Space Station for Commercial and Marketing Activities” and “Interim Policy Update for Commercial Activities on ISS”](#)). As part of that strategy, NASA plans to enable private astronaut missions (PAMs) to the ISS. These private missions must use a certified U.S. transportation vehicle that meets NASA’s ISS visiting vehicle requirements, policies, and procedures.

NASA has accommodated the pursuit of private astronaut missions under this NRA umbrella, and multiple companies have entered into this process. This focus area intends to build on the experience gathered to date and provides a path for additional entities to pursue private astronaut missions.

As a part of this announcement, NASA will identify up to two candidate private astronaut mission opportunities per year based on currently available scheduling information and resource availability. NASA is responsible for managing the mission opportunities and integrated requirements for all United States On-orbit Segment (USOS) partners, their researchers, and commercial entities. NASA is also responsible for integrating pertinent international partner and commercial vehicle provider constraints for the purposes of managing vehicle traffic to the ISS. Private astronaut mission scheduling, resource availability, and implementation will be subject to these overall opportunities and integrated requirements, and is subject to change given vehicle traffic changes, anomalies, or other unforeseen circumstances. NASA missions will take precedence.

Private Astronaut Mission (PAM) opportunities on ISS are very limited and NASA uses a process to ensure increased competition in the award of missions to the ISS. NASA has assessed general ability to support up to two missions per year, with no individual mission exceeding fourteen docked days.

PAM information and requirements [will be] consolidated into a technical library. Subsequent revisions to documents will be provided prior to contract award. Up to 2 potential Proposer representatives can request access by emailing the Contracting Officer, Ashley Chaves, at ashley.h.chaves@nasa.gov (please cc Elizabeth Kosobud at elizabeth.kosobud@nasa.gov) with the following information to be cleared for export-controlled data:

- Name, Company Name, Citizenship, Contact Address, Contact Phone Number, Contact e-mail address, Government clearance (if applicable), and NASA Domain Consolidation (NDC) Username (if applicable)
- Foreign nationals will be required to complete a non-disclosure agreement (NDA)

The due date for submission of proposals will not be extended for any offeror based on additional time necessary for obtaining access. Falsified or incorrect data may result in disqualification from selection for a PAM or future NASA solicitations.

A two-phase process for assessing and awarding PAM missions is outlined below:

Phase I (optional): Submission of a generic mission or multi-mission concept white paper to this NRA through NSPIRES:

[No Updates]

Phase II: Submission of proposals to NRA mission specific opportunity announcements:

This phase provides Proposers the ability to submit a mission-specific proposal against NASA available flight opportunities that will be competitively evaluated for a possible award. Proposals will be solicited annually, or as flight opportunities become available. Details associated with each request for mission-specific announcements will be made available through this NRA and updated as required. NASA reserves the right to cancel or alter this solicitation. Required proposal content, evaluation criteria, and schedule will be updated in this NRA, and can be found in sub-sections to Focus Area 4A (e.g. 4A.1). The sub-sections below outline the detailed instructions for submitting proposals to a Private Astronaut Mission-specific solicitation as a part of Focus Area 4A. These instructions are unique to Focus Area 4A and are different from the other Focus Areas described in this NRA.

4A Section 1.0 – Objectives

The primary objectives of Focus Area 4A are the following:

- Successfully execute a Private Astronaut Mission (PAM) to the International Space Station (ISS) with the goal of safely flying Private Astronauts to accomplish public and private research, in-space manufacturing, assembling and servicing activities, and/or commercial, marketing and utilization activities.
- To broaden and expand the capabilities of the commercial market consistent with NASA's vision for the commercialization of a LEO economy by offering the ISS as a destination for PAMs, as well as utilizing the ISS in the development of Commercial LEO Destinations.
- To allow commercial entities the opportunity to utilize the ISS through the execution of a PAM in a manner that is consistent with NASA's commercial policy on the Use of ISS for Commercial and Marketing Activities and Interim Policy Update for Commercial Activities on ISS, and as such does not reflect unfavorably on NASA, any ISS Partner, or any ISS-related entity.
- To broaden the accessibility of LEO.

Through this NRA Focus Area, NASA intends to reach an agreement for flight opportunities specified in the mission specific solicitation. Developing the commercial markets, activities, and achieving commercial viability for any PAM is the PAM Provider's responsibility, not the Government's. Financial viability of the mission should not rely on the procurement of services by NASA. Although a PAM Provider may choose to offer services of interest to NASA (such as cargo transport and cargo return) as part of a proposed mission, there is no commitment on the part of the Government to procure such services.

4A Section 2.0 – Eligibility

This Focus Area is restricted to United States industry entities. United States industry entities are defined as any corporation, partnership, joint venture, association, or other entity that meets the requirements of 51 United States Code (U.S.C.) § 50101, excluding government entities such as NASA Centers, National Laboratories, and Federally Funded Research and Development Centers.

The selected Offeror(s) shall adhere to 51 United States Code (U.S.C.) § 50131 in performance of this agreement by acquiring space transportation services from United States commercial providers. The Offeror(s) shall also comply with the National Space Transportation Policy by using U.S. manufactured launch vehicles. For a launch vehicle to qualify as a domestic end product, the cost of its components; mined, produced, or manufactured in the United States, must exceed fifty (50) percent of the cost of all of its components. The cost of each component includes transportation costs to the place of incorporation into the launch vehicle and any applicable duty (whether or not a duty-free entry certificate is issued). “Components” means those materials and supplies directly incorporated into the domestic end product.

4A Section 3.0 – Proposal Submission

Except as stated below, Proposers shall prepare proposals in accordance with instructions in Focus Area 4A of this NRA and in NFS 1852.235-72, *Instructions for Responding to NASA Research Announcements*. Additional information in this solicitation is subject to the default Agency rules in the *Guidebook for Proposals Responding to a NASA Notice of Funding Opportunity (NOFO)*. Where any discrepancies exist between the instructions in Focus Area 4A of this NRA and the *Guidebook for Proposals Responding to a NASA Notice of Funding Opportunity (NOFO)*, the information in Focus Area 4A of this NRA takes precedence.

Proposals shall be submitted electronically to NSPIRES at <https://nspires.nasaprs.com/external/>, which requires user registration. Hard copies will not be accepted. Proposals that are late will be handled in accordance with NASA’s policy as given in Section (4.1 of the *Guidebook for Proposals Responding to a NASA Notice of Funding Opportunity (NOFO)* (also see Sections 3.2), and in accordance with NFS 1815.208. Proposals received after the due date and time are considered late and will not be evaluated. Proposers may submit up to two proposals. Each proposal shall represent a single private astronaut mission concept. The government will evaluate each proposal for potential award of a flight opportunity, as defined by Focus Area 4A.1 – Solicitation for Private Astronaut Missions’ Provider for Flight Opportunity, that meets the launch window and sequence constraints provided in the proposal. Each proposal shall be submitted through NSPIRES separately.

Proposers must ensure information and certifications provided in <https://sam.gov/content/home> is current at the time of proposal submission in order to ensure all eligibility requirements outlined in 4A Section 2.0 are met.

4A Section 3.1 – Proposal Content

Proposals shall be structured as detailed in this section. Proposals shall be limited to no more than 65 pages, not including any attachments. Only attachments that are specifically requested in the content described below shall be included, and page limits for those attachments are provided in section *vii. Attachments*. Any pages submitted beyond the specified page count will not be evaluated. For specific instructions on proposal formatting not covered in these instructions, refer to section 2.6. Standard Proposal Style Formats of the *Guidebook for Proposals Responding to a NASA Notice of Funding Opportunity (NOFO)*. Proposals and attachments must be submitted in .pdf format.

Sections and content for each mission proposed shall include, at a minimum, the following material:

- i. Title Page (1 page):
 - a. Include any Notice of Restriction on Use and Disclosure of Proposal Information
 - b. The proposer and proposal name/title (Requirement is to be a US owned company)
 - c. Date of the proposal
 - d. The title, NRA number, including version associated with announcement
 - e. Organization name and address
 - f. Proposer Point of Contact name, title, e-mail address, and phone number
- ii. Proposal Abstract (not to exceed 3 pages):
 - a. Provide an overview of the proposed mission concept, including key mission parameters, objectives, teaming arrangements, as well as the Proposer's experience, capabilities, and qualifications. Proposals shall clearly define all mission objectives. Proposers are encouraged to propose multiple objectives. Proposals shall not include operational objectives that involve Private Astronauts performing Extra-Vehicular Activities (EVAs) or operating ISS Robotic Systems. NASA will not award a mission to an Offeror who proposes the performance of ISS EVAs or operation of ISS Robotic Systems as a significant mission objective.
 - b. Provide any constraints and flexibilities (including launch window and sequence constraints) in flight opportunity assignment.
 - c. Describe how proposed mission fits in with NASA's vision of LEO commercialization and the objectives of the solicitation.
- iii. Table of Contents (1 page):
 - a. Include a one-page Table of Contents that provides a guide to the organization and contents of the proposal.
- iv. Mission Overview (not to exceed twelve (12) pages):
 - a. Describe the concept of operations, details, and parameters of the proposed mission, as well as any areas of flexibility and/or constraints that have implications to mission execution or scheduling. This information should comply with requirements, processes, and terms and conditions as outlined in *SSP 51087 - ANXI, Private Astronaut Mission (PAM) Authorization, Coordination and Execution (PACE) Annex 1*, the generic *PAM Basic Ordering Agreement (BOA)* and *Mission Specific Order (MSO)*.
 - i. Information shall include the following:
 - 1. Proposed mission duration (both total duration and docked to ISS) and mission scheduling constraints
 - a. Provide any scheduling constraints, flexibility, or preferences related to mission windows. Consider "no earlier than" or "no later than" launch dates, order of execution for proposed missions, etc.
 - 2. Proposed launch and transportation vehicle
 - a. For any areas where the launch or transportation vehicle is not compliant with the requirements in SSP 51087 PACE Annex 1, provide details on the capability and associated schedule to meet the requirements in support of the proposed launch window.
 - 3. Number of primary and backup crew members, roles, and planned training

4. Significant mission objectives or activities to be performed during the mission, including the durations and scope of any proposed activities as well as any ISS resources being requested to support. Proposals shall clearly define all mission objectives. Proposers are encouraged to propose multiple objectives.
 5. Proposals shall not include operational objectives that involve Private Astronauts performing ISS Extra-Vehicular Activities (EVAs) or operating ISS Robotic Systems. NASA will not award a mission to an Offeror who proposes the performance of ISS EVAs or operation of ISS Robotic Systems as a significant mission objective.
- ii. Information should include the following:
1. Details of proposed crew members (including backups if applicable), including:
 - a. Names and background of Private Astronauts and/or non-NASA related crew selection criteria or process
 - b. Anticipated training and experience of each crew member, and roles
 2. Details of any NASA or commercially provided hardware to be utilized during the mission, as well as any planned operations
 3. Details of any research activities being performed as well as any ties to government or commercial entities (e.g. Center for the Advancement of Science in Space (CASIS))
 4. Details of any commercial ventures, offerings, partnerships, and/or activities proposed that significantly affect the execution of the mission.
 5. Details of any public affairs and/or outreach activities proposed
 6. Details of integrated media and filming activities proposed, including pre-flight, on-orbit, and post-flight activities. Activities, including media-related PAM activities, must follow the NASA guidelines and be approved regardless of contract negotiation or award; proposed activities falling outside of NASA policies will have to be negotiated prior to contract award. For documentaries, television series, and feature film products that are significant considerations of the proposal, Proposers shall provide details of the project that include:
 - a. Funding source and status of funding agreement
 - b. Broadcaster/Distributor and status of agreement with the broadcaster/distributor
 - c. Storyline (Script or Treatment)

For more information on NASA's policies of working on multimedia projects, see the following link: www.nasa.gov/multimedia/guidelines .

- b. Statement of acknowledgement and acceptance of the terms and conditions associated with the NASA PAM framework as documented in the generic *PAM Basic Ordering Agreement (BOA)* and *Mission Specific Order (MSO)*. Specific mission needs will determine the contracting mechanism that is to be used, but the terms and conditions will apply regardless of contracting mechanism selected.
- v. Mission Integration Approach (not to exceed twelve (12) pages):

- a. Describe the experience, capabilities, qualifications, resources, and key mission personnel of the Proposer and any key partners and contractors. Key personnel should be provided for both the Proposer and from contractors/partners in key areas of mission integration.
- b. Describe the approach, methods, and integration schedule/milestones for key areas of mission integration. Particular focus should be given to describing how the PAM Provider will maintain compliance and adherence to requirements, processes, and integration milestones outlined in *SSP 51087 - ANXI, Private Astronaut Mission (PAM) Authorization, Coordination and Execution (PACE) Annex I*, in the following key areas:
 - i. Transportation Vehicle/Provider Integration
 - ii. Crew Training
 - iii. Medical Certification, Support and Health Stabilization
 - iv. Crew provisions/hardware certification and flight integration
 - v. Payload Integration and Operations
- c. Describe the approach and teaming arrangements to support launch and landing services, including search and rescue.
- d. Provide a description of roles and responsibilities for the Proposer and all related entities, including primary interfaces with NASA and the transportation provider.
- e. Describe the major services, facilities, resources, and equipment required from NASA in order to execute the proposed mission. Use *SSP 50192 - Private Astronaut Resources, Interfaces, and Services (PARIS) for ISS*, as a reference to assist in the identification of potential resources needed. Include any unique or non-standard requests for services, facilities, resources, or equipment.
- f. Describe status and provide evidence of engagement and path to completion for the following key Integration Readiness Milestones as defined in PACE Annex 1, Section 3.1.2. Evidence of engagement shall demonstrate ability to meet the expected launch windows defined in Focus Area 4A.1 and associated mission integration milestones. Note: Attachment Section D provides additional page count for submittal of evidence as necessary
 - i. Reimbursable Space Act Agreement (RSAA) with Johnson Space Center's Flight Operations Directorate (FOD) for use of FOD facilities and contract(s) with NASA approved Training Vendor(s) showing ability to begin training by Integration Kickoff
 - ii. Contract with Transportation Provider with evidence that Transportation Provider can support expected launch timeframe
 - iii. CASIS Umbrella User Agreement if the PAM Provider intends to perform ISS National Lab sponsored research.
- g. Describe status of any teaming arrangements required to support the following key areas of mission integration. If a capability will be provided in-house, provide a description of internal capabilities that will be used to support the integration approach as detailed below.
 - i. Medical Certification, Support and Health Stabilization
 1. Describe plans to acquire proper medical resources, capabilities and personnel to meet pre-flight screening, health stabilization, psychological and on-orbit requirements.
 - ii. Hardware certification and flight integration

1. Describe any arrangements with NASA Vendor Partner(s) for cargo integration support, including hardware certification and crew provisions.

iii. Real Time Execution

1. Describe any pertinent teaming arrangements to support real-time mission operations with trained and certified ground control personnel as defined in PACE Annex 1, section 2.5.1.4., including USCV provider support consistent with existing ground team interfaces with the PAM USCV control center.
2. Any use of hardware or resources that are owned or managed by an International Partner (IP) require agreement from that IP. If significant mission objectives are proposed that will require use of IP-owned hardware systems or resources for mission success, describe any pertinent teaming arrangements with the IP to enable use during the mission.

vi. Mission Business Approach (not to exceed nine (9) pages):

- b. Describe the financial health of the Proposer executing the proposed mission to demonstrate the ability to accomplish mission activities for the duration of the integration and execution of the proposal. Identify the total costs of the mission, including costs of services procured from NASA as described in the current [NASA Pricing Policy](#) , as well as those from commercial suppliers and any other government partnerships required to execute the mission. Summarize the cash flow of the mission, including phasing of revenue and cost. State any assumptions for revenue from NASA as described in paragraph e.
- c. Describe the plan for obtaining required funds, including mission revenue, internal funds from other company operations, and/or external financing. Attach any evidence such as signed letters of intent from investors, if applicable.
- d. Provide a status of teaming/supplier agreements and/or partnerships with other commercial and/or government entities that will perform a role on this mission. Attach any evidence such as signed letters of intent (e.g. with transportation provider), if applicable.
- e. Provide a description of any services or commercial offerings being presented to NASA as a part of the proposal.

vii. PAM Commander Experience – Offerors must propose a PAM Commander. An alternate commander may also be proposed, per 3.1.vii.b, but an alternate Commander is not required. NASA will evaluate each Offeror’s proposed PAM Commander and alternate Commander, if applicable, based on the criteria below:

- a) PAM Commander - describe the quality of performance and effectiveness of the proposed PAM Commander’s previous experience as it relates to the experience requirements, below, as documented in Appendix F of *SSP 51087 - ANXI, Private Astronaut Mission (PAM) Authorization, Coordination and Execution (PACE) Annex 1*. (not to exceed 5 pages)
 1. In accordance with PACE Annex 1, the proposed PAM Commander must be a former NASA (U.S.) flown government astronaut
 - i. Lived and worked as an Expedition crew member (CDR or Flight Engineer) aboard the International Space Station as a long duration crewmember (30 continuous days or more) to ensure understanding of the nominal operational cadence of the USOS.

- ii. Due to the dynamic nature of the USOS operating environment, PAM Commander candidates must demonstrate prior participation in ISS USOS spaceflight operations by submitting evidence (e.g., narrative, Position Description, etc.) reflecting the last spaceflight operations position the individual held and the last date held. This may include spaceflight and/or ground support roles. Candidates whose active participation in ISS USOS spaceflight operations ended more than 5 years prior to the proposed training start date for associated flight opportunity must provide supporting rationale demonstrating the individual's current, active participation in similar, relevant spaceflight operations or provide a plan explaining how the proposed PAM Commander will get familiar with the current ISS USOS spaceflight operating environment prior to launch. Plans for PAM Commander re-familiarization must also be incorporated into the PAM Provider's proposed Project Schedule.
 2. Experience maintaining a harmonious and cohesive relationship among crewmembers and an appropriate level of mutual confidence and respect through an interactive, participative, and relationship-oriented approach, having taken into account the international and multicultural nature of the crew and mission
 3. Experience leading crewmembers to accomplish a specific goal during pre-flight, on-orbit, and, if applicable, post-flight activities
 4. Demonstrated spacecraft system and hardware expertise (e.g., Information Technology (IT) hardware, scheduling applications, hygiene systems, food systems, still photo cameras, video cameras, microphones, communication systems, etc.) during a spaceflight mission
 5. Experience fully and accurately communicating with the chain of command and with the ground teams in a timely manner during space mission activities
 6. Experience adhering to *Code of Conduct for the International Space Station Crew* (14 C.F.R. 1214.403)
 7. Experience adhering to the *Standards of Ethical Conduct for Employees of the Executive Branch* (5 C.F.R. Part 2635)
 8. Experience following direction per the chain of command during mission activities
 9. Demonstrated ability giving critical feedback when the receiver is not receptive to the message. Demonstrated ability receiving critical feedback and implementing corrective actions..
 10. Experience maintaining order, enforcing procedures, and taking effective actions in challenging and evolving operational environments during a spaceflight mission
 11. Provide details of applicable human spaceflight experience.
- b) Alternate PAM Commander Experience (Optional) – In the future, NASA may broaden its PAM Commander experience requirements under PACE Annex 1 to permit current or former CSA, ESA, or JAXA flown astronauts with U.S. Operating Segment Experience to be eligible to serve as Commander on a PAM. Broadening of this requirement is not guaranteed. Accordingly, the PAM Provider may propose an alternate PAM commander along with the supporting

information as described below. Should NASA broaden its PAM Commander experience requirements, the PAM Provider may designate its alternate Commander as its PAM Commander provided that such a swap does not negatively impact the PAM Provider's ability to meet the proposed mission integration schedule. NASA will evaluate each Offeror's proposed alternate PAM Commander based on the criteria below:

Describe the quality of performance and effectiveness of the proposed PAM Commander's previous experience as it relates to the experience requirements, below, and as documented in Appendix F of *SSP 51087 - ANXI, Private Astronaut Mission (PAM) Authorization, Coordination and Execution (PACE) Annex I*. (not to exceed 5 pages)

1. Current or former CSA, ESA, or JAXA flown astronaut with U.S. Operating Segment (USOS) experience.
 - i. Lived and worked as an Expedition crew member (CDR or Flight Engineer) aboard the International Space Station as a long duration crewmember (30 continuous days or more) to ensure understanding of the nominal operational cadence of the USOS.
 - ii. Due to the dynamic nature of the USOS operating environment, PAM Commander candidates must demonstrate current or prior participation in their respective agency's ISS USOS spaceflight operations by submitting evidence (e.g., narrative, Position Description, etc.) reflecting the last spaceflight operations position the individual held and the last date held. This may include spaceflight and/or ground support roles. Candidates whose active participation in ISS USOS spaceflight operations ended more than 5 years prior to the proposed training start date for associated flight opportunity) must provide supporting rationale demonstrating the individual's current, active participation in similar, relevant spaceflight operations or provide a plan explaining how the proposed PAM Commander will get familiar with the current ISS USOS spaceflight operating environment prior to launch. Plans for PAM Commander re-familiarization must also be incorporated into the PAM Provider's proposed Project Schedule.
2. Adherence to PAM Commander experience requirements 3.1.vii.a.2 – 3.1.vii.a.6 as written above.
3. Experience adhering to respective country's or agency's ethics laws, rules or requirements, such as, for former NASA Government Astronauts, the Standards of Ethical Conduct for Employees of the Executive Branch (5 C.F.R. Part 2635)
4. Adherence to PAM Commander experience requirements 3.1.vii.a.8 – 3.1.vii.a.11 as written above

viii. Attachments:

- a. Project Schedule

- i. Include major mission milestones, training activities and milestones, activities required to enable PAM Commander re-familiarization with the ISS USOS spaceflight operating environment, if required, vehicle integration activities and milestones, crew provisioning and hardware procurement, delivery and flight schedules, crew medical milestones, and research and commercial marketing integration activities (no page limit; however, data that is not schedule data will not be considered).
- b. Financial statements (not to exceed 20 pages)
 - i. Attach financial statements (cash flow, balance sheet, and income statement) for the last three years of your company and supplementary documents (as needed) covering the period of execution.
- c. Evidence of revenue and sources of funds (not to exceed 10 pages)
- d. Evidence of planned suppliers, teaming agreements, etc. (not to exceed 10 pages)
- e. PAM Budget Table (no page limit)

4A Section 4.0 – Proposal Evaluation Information

Except as stated below, proposals will be evaluated in accordance with the *NASA FAR Supplement, Section 1852.235-72*, and as reflected in the *Guidebook for Proposals Responding to a NASA Notice of Funding Opportunity (NOFO)*. All information needed to apply to this solicitation is contained in Focus Area 4A of this NRA and anything not mentioned here is subject to the default Agency rules in the *Guidebook for Proposals Responding to a NASA Notice of Funding Opportunity (NOFO)*. Proposers are responsible for understanding and complying with its procedures for the successful, timely preparation and submission of their proposals.

4A Section 4.1 – Proposal Compliance

NASA will screen all proposals for compliance with requirements of this solicitation. This includes submission of a proposal that is compliant with all of the following:

- The proposal is relevant to the objectives and/or provisions of Focus Area 4A of this NRA;
- The proposal meets the requirements for proposal format, content, and organization as specified in the *Guidebook for Proposals Responding to a NASA Notice of Funding Opportunity (NOFO)* and/or or Focus Area 4A of this NRA;
- The proposal is submitted by the submission due date and time, and be valid for 365 days;
- The proposal consists of files in Acrobat Adobe .pdf.

Non-compliant proposals may be withdrawn from the review process and declined without further review. Compliant proposals submitted in response to this NRA will undergo a full evaluation.

4A Section 4.2 – Evaluation Criteria

The evaluation criteria in this section are tailored to Focus Area 4A and will be used for the evaluation of Private Astronaut Mission specific proposals. Any materials provided in the proposal and/or the attachments may be evaluated against the evaluation criteria. Proposals that have met the requirements under Proposal Compliance

in 4A Section 4.1 will be further evaluated against three factors, 1. Merit, 2. NASA Supportability, and 3. Relevance. Factors 1 and 2 (Merit and Supportability) are of equal importance, and each are more important than Factor 3 (Relevance).

Factor 1: Merit will be evaluated based on the criteria below:

- The feasibility of executing the proposed mission within NASA's outlined framework, and ability to meet NASA's requirements in *SSP 51087 PACE Annex 1* in support of the proposed launch windows.
- The ability of the Proposer to provide the personnel, resources, financing, and partnerships necessary to plan and execute the proposed mission.
 - o Note: Mission proposals should be independently financially viable, and not require procurement of services by NASA.
- The ability of the Proposer to secure the partnerships and teaming arrangements necessary to meet the milestones to support the proposed schedule. (reference sections 3.1.v.f and 3.1.v.g)
- The reasonableness and feasibility of supporting the proposed project schedule and integration milestones.
- Quality of performance and effectiveness of the PAM Commander to meet the experience requirements as defined in *SSP 51087 PACE Annex 1* and the *Basic Ordering Agreement*. The Government can use both data provided by the offeror and data obtained from other sources in its assessment. The Government plans to use performance information obtained by the evaluation team based on data independently obtained by the Government, including, but not limited to internal NASA information related to the proposed PAM Commander's performance quality and effectiveness.
 - o If applicable, the PAM Provider's plan for ensuring the proposed PAM Commander's refamiliarization with the USOS spaceflight operating environment will be evaluated for reasonableness, feasibility, and completeness.

Factor 2: NASA Supportability will be evaluated based on the criteria listed below:

- Evaluation of risk, including safety considerations for personnel, vehicle, hardware, ground infrastructure, mission objectives and operations.
- Evaluation of NASA's ability to provide the resources, facilities, hardware, services and other items necessary to accomplish the mission.
- Evaluation of NASA's ability to support the proposed integration timeline and support the proposed mission flight objectives.
- Evaluation of impact to availability of the transportation and launch vehicles for other NASA missions.

Factor 3: Relevance will be evaluated based on the proposal's potential contribution to NASA's mission and the objectives stated in 4A Section 1.0 Objectives.

Note: If, at any time during evaluation a proposal is found to be not relevant, further evaluation will be handled in accordance with 4A Section 4.1.

The adjectival ratings listed in Appendix D of the *Guidebook for Proposals Responding to a NASA Notice of Funding Opportunity (NOFO)* will be used to evaluate Factors 1-3.

4A Section 5.0 – Selection and Award

The Government reserves the right to award with or without discussions. Per Section 3.4 of the NRA Guidebook, updates to the proposal may be requested after discussions. The offeror's initial proposal should contain the offeror's best terms. The Selection Official will select proposals as judged against the evaluation criteria, the objectives of the NOFO, programmatic considerations, and available financial resources. A higher evaluated proposal may not be selected for negotiation and/or award due to budget limitations and/or programmatic considerations.

NASA reserves the right to select for negotiation two, one, or none of the proposals received in response to this solicitation. The overall number of selections for negotiation will be dependent upon the evaluation criteria in 4A Section 4.2, the flight opportunities available, the flight planning and scheduling constraints provided in the proposal, and programmatic considerations. Once an offeror has been selected for negotiation, NASA will determine the appropriate contract or agreement mechanisms based on the Proposer's mission specific proposal. A generic Basic Ordering Agreement (BOA) and Mission Specific Order (MSO) agreement have been provided in the technical library as an example but a different agreement mechanism may be utilized. It is NASA's intent not to negotiate the terms and conditions in the BOA. If Proposer and government cannot agree to the terms and conditions or any mission specific details, or the offeror significantly changes their mission scope after selection for negotiations, the Government reserves the right to select another proposal. Selection for negotiation does not guarantee selection for award.